

In the Name of God

Computational Fluid Dynamics II

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Chapter 1

Classification of Governing Equations and Transformation

فصل ۱

دسته‌بندی معادلات حاکم و انتقال

دسته‌بندی معادلات حاکم و انتقال

- معادلات حاکم

- معادله بقای جرم

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{V}) = 0 \quad \text{Conservative form}$$

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0 \quad \text{3D Cartesian coordinates}$$

$$\frac{\partial \rho}{\partial t} + \vec{V} \cdot \vec{\nabla} \rho + \rho \vec{\nabla} \cdot \vec{V} = 0 \quad \text{Non-conservative (or primitive) form}$$

$$\frac{D\rho}{Dt} + \rho \vec{\nabla} \cdot \vec{V} = 0$$

$$\frac{D()}{Dt} = \frac{\partial()}{\partial t} + \vec{V} \cdot \vec{\nabla}() \equiv \text{Material derivative}$$

- تعریف مشتق مادی

دسته‌بندی معادلات حاکم و انتقال

- معادلات حاکم

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{V}) = 0 \quad \& \quad \frac{D\rho}{Dt} + \rho \vec{\nabla} \cdot \vec{V} = 0$$

- معادله بقای جرم

- برای جریان تراکم‌ناپذیر

$$\frac{D\rho}{Dt} = 0 \quad \rightarrow \quad \vec{\nabla} \cdot \vec{V} = \frac{1}{\rho} \frac{D\rho}{Dt} \equiv \text{Dilatation} = 0$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad \text{3D Cartesian coordinates}$$

$$\vec{\nabla} \times \vec{V} = 0 \quad \rightarrow \quad \vec{V} = \vec{\nabla} \phi$$

- برای جریان غیرچرخشی

$$\vec{\nabla} \cdot \vec{V} = \vec{\nabla} \cdot \vec{\nabla} \phi = \nabla^2 \phi = 0 \quad \text{Laplace equation for potential function}$$

دسته‌بندی معادلات حاکم و انتقال

• معادلات حاکم

- معادله بقای ممنتوم

$$\rho \frac{D\vec{V}}{Dt} = \rho \left(\frac{\partial \vec{V}}{\partial t} + \vec{V} \cdot \vec{\nabla} \vec{V} \right) = \vec{\nabla} \cdot \underline{\underline{\sigma}} + \rho \vec{f} \quad \text{Non-conservative form}$$

$$\underline{\underline{\sigma}} = -p \underline{\underline{I}} + \underline{\underline{\tau}} \quad \& \quad \underline{\underline{\tau}} = 2\mu \underline{\underline{\varepsilon}} + \lambda \vec{\nabla} \cdot \vec{V} \underline{\underline{I}} \quad \& \quad \underline{\underline{\varepsilon}} = \frac{1}{2} \left(\vec{\nabla} \vec{V} + (\vec{\nabla} \vec{V})^T \right)$$

- تعریف فشار مکانیکی

$$\bar{p} = -\frac{1}{3} \sigma_{ii} = -\frac{1}{3} (\tau_{11} + \tau_{22} + \tau_{33}) = p - \left(\lambda + \frac{2}{3} \mu \right) \vec{\nabla} \cdot \vec{V}$$

$$\lambda + \frac{2}{3} \mu = 0 \quad \text{Stokes' hypothesis} \quad \rightarrow \quad \underbrace{\bar{p}}_{\text{mechanical pressure}} = \underbrace{p}_{\text{thermodynamic pressure}}$$

دسته‌بندی معادلات حاکم و انتقال

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- معادله بقای ممنتوم

$$\rho \left(\frac{\partial \vec{V}}{\partial t} + \vec{V} \cdot \vec{\nabla} \vec{V} \right) = \vec{\nabla} \cdot \underline{\underline{\sigma}} + \rho \vec{f} \quad \& \quad \vec{V} \left(\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{V}) \right) = 0$$

$$\frac{\partial (\rho \vec{V})}{\partial t} + \vec{\nabla} \cdot (\rho \vec{V} \vec{V}) = \vec{\nabla} \cdot \underline{\underline{\sigma}} + \rho \vec{f} \quad \text{Conservative form}$$

$$\underline{\underline{\sigma}} = -p \underline{\underline{I}} + \underline{\underline{\tau}} \quad \& \quad \underline{\underline{\tau}} = 2\mu \underline{\underline{\varepsilon}} + \lambda \vec{\nabla} \cdot \vec{V} \underline{\underline{I}} \quad \& \quad \underline{\underline{\varepsilon}} = \frac{1}{2} \left(\vec{\nabla} \vec{V} + (\vec{\nabla} \vec{V})^T \right)$$

$$\begin{cases} \tau_{xx} = 2\mu \frac{\partial u}{\partial x} + \lambda \vec{\nabla} \cdot \vec{V} \\ \tau_{yy} = 2\mu \frac{\partial v}{\partial y} + \lambda \vec{\nabla} \cdot \vec{V} \\ \tau_{zz} = 2\mu \frac{\partial w}{\partial z} + \lambda \vec{\nabla} \cdot \vec{V} \end{cases} \quad \& \quad \begin{cases} \tau_{xy} = \tau_{yx} = \mu \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) \\ \tau_{xz} = \tau_{zx} = \mu \left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \right) \\ \tau_{yz} = \tau_{zy} = \mu \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right) \end{cases}$$

دسته‌بندی معادلات حاکم و انتقال

• معادلات حاکم

- معادله بقای ممنتوم

$$\frac{\partial(\rho \vec{V})}{\partial t} + \vec{\nabla} \cdot (\rho \vec{V} \vec{V}) = -\vec{\nabla} p + \vec{\nabla} \cdot \underline{\underline{\tau}} + \rho \vec{f}$$

x - mom.:

$$\frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho u^2)}{\partial x} + \frac{\partial(\rho uv)}{\partial y} + \frac{\partial(\rho uw)}{\partial z} = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} + \rho f_x$$

$$\begin{aligned} \frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho u^2 + p)}{\partial x} + \frac{\partial(\rho uv)}{\partial y} + \frac{\partial(\rho uw)}{\partial z} &= \frac{\partial}{\partial x} \left(2\mu \frac{\partial u}{\partial x} + \lambda \vec{\nabla} \cdot \vec{V} \right) \\ &+ \frac{\partial}{\partial y} \left(\mu \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) \right) + \frac{\partial}{\partial z} \left(\mu \left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \right) \right) + \rho f_x \end{aligned}$$

دسته‌بندی معادلات حاکم و انتقال

- معادلات حاکم

- معادله بقای ممنتوم

- برای جریان تراکم‌ناپذیر

$$\frac{\partial(\rho \vec{V})}{\partial t} + \vec{\nabla} \cdot (\rho \vec{V} \vec{V}) = -\vec{\nabla} p + \vec{\nabla} \cdot \underline{\underline{\tau}} + \rho \vec{f}$$

$$\rho \left(\frac{\partial \vec{V}}{\partial t} + \vec{\nabla} \cdot (\vec{V} \vec{V}) \right) = -\vec{\nabla} p + \vec{\nabla} \cdot \underline{\underline{\tau}} + \rho \vec{f}$$

$$x - \text{mom.:} \quad \rho \left\{ \frac{\partial(u)}{\partial t} + \frac{\partial(u^2)}{\partial x} + \frac{\partial(uv)}{\partial y} + \frac{\partial(uw)}{\partial z} \right\} = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} + \rho f_x$$

$$\begin{aligned} \rho \left(\frac{\partial(u)}{\partial t} + \frac{\partial(u^2 + p)}{\partial x} + \frac{\partial(uv)}{\partial y} + \frac{\partial(uw)}{\partial z} \right) &= \frac{\partial}{\partial x} \left(2\mu \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial y} \left(\mu \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) \right) + \frac{\partial}{\partial z} \left(\mu \left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \right) \right) + \rho f_x \\ &= \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) + \rho f_x = \mu \nabla^2 u + \rho f_x \end{aligned}$$

$$\rho \left(\frac{\partial \vec{V}}{\partial t} + \vec{\nabla} \cdot (\vec{V} \vec{V}) \right) = -\vec{\nabla} p + \mu \nabla^2 \vec{V} + \rho \vec{f}$$

دسته‌بندی معادلات حاکم و انتقال

• معادلات حاکم

- معادله بقای انرژی

- فرم انرژی کل

$$E_t = \rho \left(e + \frac{1}{2} \vec{V} \cdot \vec{V} \right) = \text{Total energy per unit volume}$$

$$\frac{DE_t}{Dt} = \frac{\partial E_t}{\partial t} + \vec{V} \cdot \vec{\nabla} E_t = -\vec{\nabla} \cdot \vec{q} - \vec{\nabla} \cdot (p\vec{V}) + \vec{\nabla} \cdot (\underline{\underline{\tau}} \cdot \vec{V}) + \rho \vec{f} \cdot \vec{V}$$

$$\frac{\partial E_t}{\partial t} + \vec{\nabla} \cdot (E_t \vec{V}) = -\vec{\nabla} \cdot \vec{q} - \vec{\nabla} \cdot (p\vec{V}) + \vec{\nabla} \cdot (\underline{\underline{\tau}} \cdot \vec{V}) \quad \text{Conservative form}$$

$$\frac{\partial E_t}{\partial t} + \vec{\nabla} \cdot [(E_t + p)\vec{V}] = -\vec{\nabla} \cdot \vec{q} + \vec{\nabla} \cdot (\underline{\underline{\tau}} \cdot \vec{V})$$

- فرم انرژی داخلی

$$\rho \frac{De}{Dt} = \rho \left(\frac{\partial e}{\partial t} + \vec{V} \cdot \vec{\nabla} e \right) = -\vec{\nabla} \cdot \vec{q} - p \vec{\nabla} \cdot \vec{V} + \underbrace{\vec{\nabla} \cdot (\underline{\underline{\tau}} \cdot \vec{V}) - \vec{V} \cdot (\vec{\nabla} \cdot \underline{\underline{\tau}})}_{\Phi = \text{Viscous dissipation} = \underline{\underline{\tau}} : \underline{\underline{\varepsilon}}}$$

دسته‌بندی معادلات حاکم و انتقال

- معادلات حاکم

- معادله بقای انرژی

- فرم آنتالپی استاتیکی

$$h = e + \frac{p}{\rho} = \text{Static enthalpy}$$

$$\rho \frac{Dh}{Dt} = \rho \left(\frac{\partial h}{\partial t} + \vec{V} \cdot \vec{\nabla} h \right) = -\vec{\nabla} \cdot \vec{q} + \frac{Dp}{Dt} + \Phi$$

- فرم آنتالپی کل

$$h_0 = h + \frac{1}{2} \vec{V} \cdot \vec{V} = e + \frac{p}{\rho} + \frac{1}{2} \vec{V} \cdot \vec{V} = \text{Total enthalpy}$$

$$\rho \frac{Dh_0}{Dt} = \rho \left(\frac{\partial h_0}{\partial t} + \vec{V} \cdot \vec{\nabla} h_0 \right) = -\vec{\nabla} \cdot \vec{q} + \frac{\partial p}{\partial t} + \vec{\nabla} \cdot (\underline{\underline{\tau}} \cdot \vec{V}) + \rho \vec{f} \cdot \vec{V}$$

- قانون هدایت فوریه

$$\text{Fourier's law : } \vec{q} = -k \vec{\nabla} T$$

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• معادلات حاکم

- معادله بقای انرژی

- برای جریان تراکم‌ناپذیر

$$\rho \frac{De}{Dt} = \rho \left(\frac{\partial e}{\partial t} + \vec{V} \cdot \vec{\nabla} e \right) = -\vec{\nabla} \cdot \vec{q} - p \vec{\nabla} \cdot \vec{V} + \Phi$$

$$\rho \frac{De}{Dt} = \rho \left(\frac{\partial e}{\partial t} + \vec{V} \cdot \vec{\nabla} e \right) = -\vec{\nabla} \cdot \vec{q} + \Phi$$

$$\rho \frac{De}{Dt} = \rho \left(\frac{\partial e}{\partial t} + \vec{V} \cdot \vec{\nabla} e \right) = -\vec{\nabla} \cdot (-k \vec{\nabla} T) + \Phi$$

$$\rho \frac{De}{Dt} = \rho \left(\frac{\partial e}{\partial t} + \vec{V} \cdot \vec{\nabla} e \right) = k \nabla^2 T + \Phi \quad \text{for } k = \text{const.}$$

$$\rho \frac{De}{Dt} = \rho \left(\frac{\partial e}{\partial t} + u \frac{\partial e}{\partial x} + v \frac{\partial e}{\partial y} + w \frac{\partial e}{\partial z} \right) = k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) + \Phi \quad \text{3D Cartesian coordinates}$$

$$\Phi = \vec{\nabla} \cdot (\underline{\underline{\tau}} \cdot \vec{V}) - \vec{V} \cdot (\vec{\nabla} \cdot \underline{\underline{\tau}}) = \underline{\underline{\tau}} : \underline{\underline{\varepsilon}} = 2\mu \underline{\underline{\varepsilon}} : \underline{\underline{\varepsilon}}$$

$$= \mu \left\{ 2 \left(\frac{\partial u}{\partial x} \right)^2 + 2 \left(\frac{\partial v}{\partial y} \right)^2 + 2 \left(\frac{\partial w}{\partial z} \right)^2 + \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right)^2 + \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} \right)^2 + \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right)^2 \right\}$$

دسته‌بندی معادلات حاکم و انتقال

- معادلات حاکم

For homogenous equilibrium gas: $e = e(\rho, T), h = h(p, T)$ - خواص ترمودینامیکی

$$\frac{De}{Dt} = \left(\frac{\partial e}{\partial T} \right)_{\rho} \frac{DT}{Dt} + \left(\frac{\partial e}{\partial \rho} \right)_T \frac{D\rho}{Dt} \quad \& \quad c_v = \left(\frac{\partial e}{\partial T} \right)_{\rho} = \text{Specific heat at constant volume}$$

$$\frac{Dh}{Dt} = \left(\frac{\partial h}{\partial T} \right)_p \frac{DT}{Dt} + \left(\frac{\partial h}{\partial p} \right)_T \frac{Dp}{Dt} \quad \& \quad c_p = \left(\frac{\partial h}{\partial T} \right)_p = \text{Specific heat at constant pressure}$$

For themally perfect (not chemically reacting) gas: $e = e(T), h = h(T)$

$$\frac{De}{Dt} = c_v \frac{DT}{Dt} \quad \& \quad de = c_v dT$$

$$\frac{Dh}{Dt} = c_p \frac{DT}{Dt} \quad \& \quad dh = c_p dT$$

دسته‌بندی معادلات حاکم و انتقال

• معادلات حاکم

- خواص ترمودینامیکی

For calorically perfect gas: $c_v = \text{const.}$ & $c_p = \text{const.}$

$$e = c_v T \quad \& \quad h = c_p T$$

$$\begin{cases} \frac{c_p}{c_v} = \gamma \\ c_p - c_v = R \end{cases} \rightarrow \begin{cases} c_p = \frac{\gamma R}{\gamma - 1} \\ c_v = \frac{R}{\gamma - 1} \end{cases} \quad \& \quad R = 287.05 \frac{J}{\text{kg K}} \quad \text{for air at standard conditions}$$

$$p = \rho R T = \rho R \frac{e}{c_v} = \rho e (\gamma - 1) \quad \& \quad E_t = \rho \left(e + \frac{1}{2} V^2 \right) = \frac{p}{\gamma - 1} + \frac{1}{2} \rho V^2$$

Sutherland's law :

$$\mu = C_1 \frac{T^{3/2}}{T + C_2} \quad \& \quad C_1 = 1.458 \times 10^{-6} \frac{\text{kg}}{\text{m} \cdot \text{s} \cdot \text{K}^{1/2}} \quad \& \quad C_2 = 110.4 \text{ K}$$

$$k = C_3 \frac{T^{3/2}}{T + C_4} \quad \& \quad C_3 = 2.495 \times 10^{-3} \frac{\text{kg} \cdot \text{m}}{\text{s}^3 \cdot \text{K}^{3/2}} \quad \& \quad C_4 = 194 \text{ K}$$

دسته‌بندی معادلات حاکم و انتقال

- معادلات حاکم

- خواص ترمودینامیکی

Sutherland's law :

$$\frac{\mu}{\mu_0} = \left(\frac{T}{T_0} \right)^{3/2} \frac{T_0 + S}{T + S} \quad \& \quad T_0 = 273 \text{ K} \quad \& \quad S = 110.4 \text{ K}$$

Prandtl number (Pr) :

$$\text{Pr} = \frac{\nu}{\alpha} = \frac{\mu / \rho}{k / \rho c_p} = \frac{\mu c_p}{k} = \text{const.} \quad \rightarrow \quad k = \frac{\mu c_p}{\text{Pr}}$$

$$\frac{c_p}{\text{Pr}} = \text{const.} \quad \text{for most gases}$$

$$\text{Pr} = 0.72 \quad \text{for air}$$

دسته‌بندی معادلات حاکم و انتقال

• معادلات ناویر-استوکس تراکم‌پذیر دوبعدی

- فرض گاز کامل

$$\frac{\partial Q}{\partial t} + \frac{\partial F_i}{\partial x} + \frac{\partial G_i}{\partial y} = \frac{\partial F_v}{\partial x} + \frac{\partial G_v}{\partial y}$$

$$Q = \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ E_t \end{pmatrix}, \quad F_i = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ (E_t + p)u \end{pmatrix}, \quad G_i = \begin{pmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ (E_t + p)v \end{pmatrix}, \quad F_v = \begin{pmatrix} 0 \\ \tau_{xx} \\ \tau_{xy} \\ u\tau_{xx} + v\tau_{xy} - q_x \end{pmatrix}, \quad G_v = \begin{pmatrix} 0 \\ \tau_{xy} \\ \tau_{yy} \\ u\tau_{xy} + v\tau_{yy} - q_y \end{pmatrix}$$

$$E_t = \rho \left(e + \frac{1}{2} V^2 \right) = \frac{p}{\gamma - 1} + \frac{1}{2} \rho V^2 \quad \& \quad V^2 = u^2 + v^2$$

$$\tau_{xx} = (2\mu + \lambda) \frac{\partial u}{\partial x} + \lambda \frac{\partial v}{\partial y} \quad \& \quad \tau_{yy} = (2\mu + \lambda) \frac{\partial v}{\partial y} + \lambda \frac{\partial u}{\partial x} \quad \& \quad \tau_{xy} = \tau_{yx} = \mu \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right)$$

$$q_x = -k \frac{\partial T}{\partial x} \quad \& \quad q_y = -k \frac{\partial T}{\partial y}$$

دسته‌بندی معادلات حاکم و انتقال

- معادلات ناویر-استوکس تراکم‌پذیر تقارن محوری

- فرض گاز کامل

$$\frac{\partial Q}{\partial t} + \frac{\partial F_i}{\partial x} + \frac{\partial G_i}{\partial y} + \frac{j}{y}(G_i + H_i) = \frac{\partial F_v}{\partial x} + \frac{\partial G_v}{\partial y} + \frac{j}{y}(G_v + H_v)$$

$$H_i = \begin{pmatrix} 0 \\ 0 \\ -p \\ 0 \end{pmatrix}, \quad H_v = \begin{pmatrix} 0 \\ 0 \\ \tau_{\phi\phi} \\ 0 \end{pmatrix}$$

$$\tau_{xx} = \tau_{xx}|_{2D} + \frac{j}{y}\lambda v \quad \& \quad \tau_{yy} = \tau_{yy}|_{2D} + \frac{j}{y}\lambda v \quad \& \quad \tau_{\phi\phi} = (2\mu + \lambda)\frac{v}{y} + \lambda\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right)$$

$$j = \begin{cases} 0 & \text{Two-dimension flow} \\ 1 & \text{Axisymmetric flow} \end{cases}$$

دسته‌بندی معادلات حاکم و انتقال

Set 1:

• بی‌بعدسازی معادلات حاکم

$$\tilde{t} = \frac{t}{L / u_{\infty}}, \quad \tilde{x} = \frac{x}{L}, \quad \tilde{y} = \frac{y}{L}, \quad \tilde{u} = \frac{u}{u_{\infty}}, \quad \tilde{v} = \frac{v}{u_{\infty}},$$

- پارامترهای بی‌بعد

$$\tilde{\rho} = \frac{\rho}{\rho_{\infty}}, \quad \tilde{T} = \frac{T}{T_{\infty}}, \quad \tilde{p} = \frac{p}{\rho_{\infty} u_{\infty}^2}, \quad \tilde{E}_t = \frac{E_t}{\rho_{\infty} u_{\infty}^2},$$

$$\tilde{\mu} = \frac{\mu}{\mu_{\infty}}, \quad \tilde{k} = \frac{k}{k_{\infty}}$$

Set 2:

$$\tilde{t} = \frac{t}{L / (a_{\infty} / \sqrt{\gamma})}, \quad \tilde{x} = \frac{x}{L}, \quad \tilde{y} = \frac{y}{L}, \quad \tilde{u} = \frac{u}{a_{\infty} / \sqrt{\gamma}}, \quad \tilde{v} = \frac{v}{a_{\infty} / \sqrt{\gamma}},$$

$$\tilde{\rho} = \frac{\rho}{\rho_{\infty}}, \quad \tilde{T} = \frac{T}{T_{\infty}}, \quad \tilde{p} = \frac{p}{p_{\infty}}, \quad \tilde{E}_t = \frac{E_t}{p_{\infty}},$$

$$\tilde{\mu} = \frac{\mu}{\mu_{\infty}}, \quad \tilde{k} = \frac{k}{k_{\infty}}$$

دسته‌بندی معادلات حاکم و انتقال

• فرم بی‌بعد معادلات ناویر-استوکس تراکم‌پذیر دوبعدی

$$\frac{\partial Q}{\partial t} + \frac{\partial F_i}{\partial x} + \frac{\partial G_i}{\partial y} = \frac{1}{\text{Re}} \left(\frac{\partial F_v}{\partial x} + \frac{\partial G_v}{\partial y} \right)$$

- استفاده از پارامترهای بی‌بعد Set 1

$$Q = \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ E_t \end{pmatrix}, F_i = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ (E_t + p)u \end{pmatrix}, G_i = \begin{pmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ (E_t + p)v \end{pmatrix}, F_v = \begin{pmatrix} 0 \\ \tau_{xx} \\ \tau_{xy} \\ u\tau_{xx} + v\tau_{xy} + \frac{\gamma}{\text{Pr}} k \frac{\partial e}{\partial x} \end{pmatrix}, G_v = \begin{pmatrix} 0 \\ \tau_{xy} \\ \tau_{yy} \\ u\tau_{xy} + v\tau_{yy} + \frac{\gamma}{\text{Pr}} k \frac{\partial e}{\partial y} \end{pmatrix}$$

$$E_t = \rho \left(e + \frac{1}{2} V^2 \right) = \frac{p}{\gamma - 1} + \frac{1}{2} \rho V^2 \quad \& \quad p = \frac{1}{\gamma M_\infty^2} \rho T \quad \& \quad k = \mu$$

$$\text{Re} = \text{Re}_\infty = \frac{\rho_\infty u_\infty L}{\mu_\infty} \quad \& \quad M_\infty = \frac{V_\infty}{a_\infty} = \frac{V_\infty}{\sqrt{\gamma R T_\infty}}$$

دسته‌بندی معادلات حاکم و انتقال

• فرم بی‌بعد معادلات ناویر-استوکس تراکم‌پذیر دوبعدی

$$\frac{\partial Q}{\partial t} + \frac{\partial F_i}{\partial x} + \frac{\partial G_i}{\partial y} = \frac{1}{\text{Re}} \left(\frac{\partial F_v}{\partial x} + \frac{\partial G_v}{\partial x} \right)$$

- استفاده از پارامترهای بی‌بعد Set 2

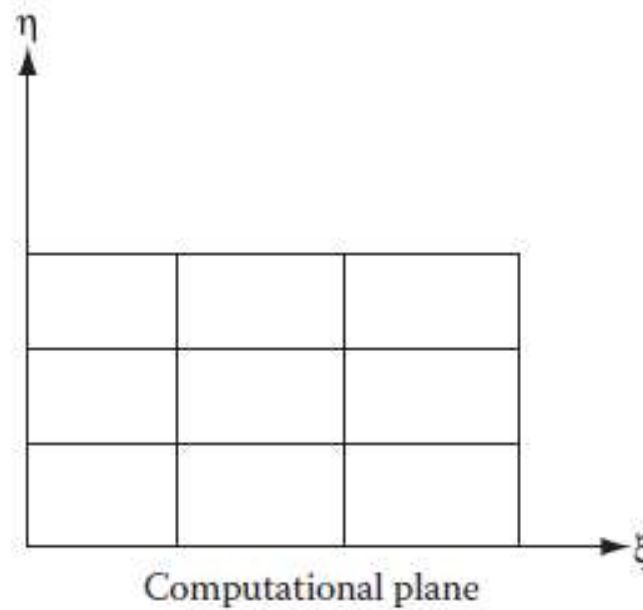
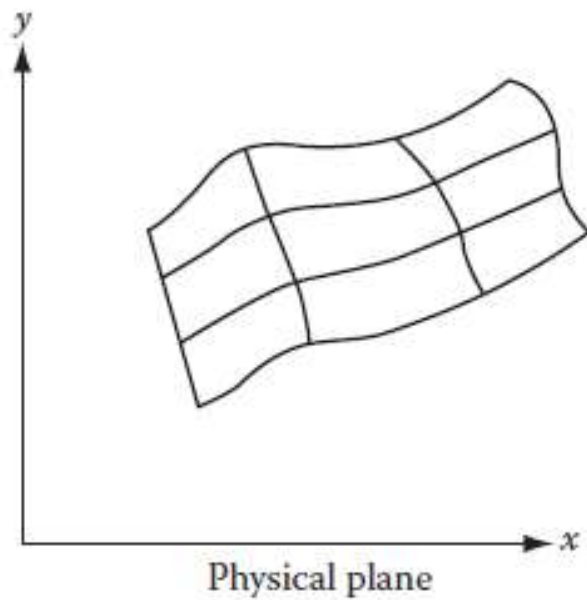
$$Q = \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ E_t \end{pmatrix}, F_i = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ (E_t + p)u \end{pmatrix}, G_i = \begin{pmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ (E_t + p)v \end{pmatrix}, F_v = \begin{pmatrix} 0 \\ \tau_{xx} \\ \tau_{xy} \\ u\tau_{xx} + v\tau_{xy} + \frac{\gamma}{\text{Pr}} k \frac{\partial e}{\partial x} \end{pmatrix}, G_v = \begin{pmatrix} 0 \\ \tau_{xy} \\ \tau_{yy} \\ u\tau_{xy} + v\tau_{yy} + \frac{\gamma}{\text{Pr}} k \frac{\partial e}{\partial y} \end{pmatrix}$$

$$E_t = \rho \left(e + \frac{1}{2} V^2 \right) = \frac{p}{\gamma - 1} + \frac{1}{2} \rho V^2 \quad \& \quad p = \rho T \quad \& \quad k = \mu$$

$$\text{Re} = \frac{\text{Re}_\infty}{\sqrt{\gamma}} \frac{1}{M_\infty} \quad \& \quad \text{Re}_\infty = \frac{\rho_\infty u_\infty L}{\mu_\infty}$$

دسته‌بندی معادلات حاکم و انتقال

- انتقال مختصات (شبکه دوبعدی و غیرمتحرک)



$$\begin{cases} \xi = \xi(x, y) \\ \eta = \eta(x, y) \end{cases}$$

&

$$\begin{cases} x = x(\xi, \eta) \\ y = y(\xi, \eta) \end{cases}$$

دسته‌بندی معادلات حاکم و انتقال

- انتقال مختصات (شبکه دوبعدی و غیرمتحرک)

- مشتق زنجیره‌ای

$$\begin{cases} \xi = \xi(x, y) \\ \eta = \eta(x, y) \end{cases} \rightarrow \begin{cases} \frac{\partial}{\partial x} = \frac{\partial \xi}{\partial x} \frac{\partial}{\partial \xi} + \frac{\partial \eta}{\partial x} \frac{\partial}{\partial \eta} = \xi_x \frac{\partial}{\partial \xi} + \eta_x \frac{\partial}{\partial \eta} \\ \frac{\partial}{\partial y} = \frac{\partial \xi}{\partial y} \frac{\partial}{\partial \xi} + \frac{\partial \eta}{\partial y} \frac{\partial}{\partial \eta} = \xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta} \end{cases}$$

- متریک‌های تبدیل

$$\begin{cases} x = x(\xi, \eta) \\ y = y(\xi, \eta) \end{cases} \rightarrow \begin{cases} \frac{\partial}{\partial \xi} = \frac{\partial x}{\partial \xi} \frac{\partial}{\partial x} + \frac{\partial y}{\partial \xi} \frac{\partial}{\partial y} = x_\xi \frac{\partial}{\partial x} + y_\xi \frac{\partial}{\partial y} \\ \frac{\partial}{\partial \eta} = \frac{\partial x}{\partial \eta} \frac{\partial}{\partial x} + \frac{\partial y}{\partial \eta} \frac{\partial}{\partial y} = x_\eta \frac{\partial}{\partial x} + y_\eta \frac{\partial}{\partial y} \end{cases}$$

$$\begin{pmatrix} \partial_x \\ \partial_y \end{pmatrix} = \begin{pmatrix} \xi_x & \eta_x \\ \xi_y & \eta_y \end{pmatrix} \begin{pmatrix} \partial_\xi \\ \partial_\eta \end{pmatrix} \quad \& \quad \begin{pmatrix} \partial_\xi \\ \partial_\eta \end{pmatrix} = \begin{pmatrix} x_\xi & y_\xi \\ x_\eta & y_\eta \end{pmatrix} \begin{pmatrix} \partial_x \\ \partial_y \end{pmatrix}$$

دسته‌بندی معادلات حاکم و انتقال

- انتقال مختصات (شبکه دوبعدی و غیرمتحرک)

$$\begin{pmatrix} \xi_x & \xi_y \\ \eta_x & \eta_y \end{pmatrix} = \begin{pmatrix} x_\xi & x_\eta \\ y_\xi & y_\eta \end{pmatrix}^{-1} = J \begin{pmatrix} y_\eta & -x_\eta \\ -y_\xi & x_\xi \end{pmatrix}$$

- متریک‌های تبدیل

$$\xi_x = J y_\eta \quad \& \quad \eta_x = -J y_\xi \quad \& \quad \xi_y = -J x_\eta \quad \& \quad \eta_y = J x_\xi$$

- ژاکوبین تبدیل

$$\xi_x = \frac{\partial \xi}{\partial x} \approx \frac{\Delta \xi}{\Delta x} = \frac{\text{length in } \xi - \eta \text{ plane}}{\text{length in } x - y \text{ plane}}$$

$$J = \frac{\text{area of computational space}}{\text{area of physical space}} = \frac{\partial(\xi, \eta)}{\partial(x, y)} = \begin{vmatrix} \xi_x & \xi_y \\ \eta_x & \eta_y \end{vmatrix}$$

$$= \frac{1}{J^{-1}} = \frac{1}{\frac{\partial(x, y)}{\partial(\xi, \eta)}} = \frac{1}{\begin{vmatrix} x_\xi & x_\eta \\ y_\xi & y_\eta \end{vmatrix}} = \frac{1}{x_\xi y_\eta - x_\eta y_\xi}$$

دسته‌بندی معادلات حاکم و انتقال

$$\begin{bmatrix} \xi_x & \xi_y & \xi_z \\ \eta_x & \eta_y & \eta_z \\ \zeta_x & \zeta_y & \zeta_z \end{bmatrix} = \begin{bmatrix} x_\xi & x_\eta & x_\zeta \\ y_\xi & y_\eta & y_\zeta \\ z_\xi & z_\eta & z_\zeta \end{bmatrix}^{-1}$$

• انتقال مختصات (شبکه سه‌بعدی و غیرمتحرک)

- متریک‌های تبدیل

$$\begin{aligned} \xi_x &= J(y_\eta z_\zeta - y_\zeta z_\eta), & \xi_y &= J(x_\zeta z_\eta - x_\eta z_\zeta), & \xi_z &= J(x_\eta y_\zeta - x_\zeta y_\eta) \\ \eta_x &= J(y_\zeta z_\xi - y_\xi z_\zeta), & \eta_y &= J(x_\xi z_\zeta - x_\zeta z_\xi), & \eta_z &= J(x_\zeta y_\xi - x_\xi y_\zeta) \\ \zeta_x &= J(y_\xi z_\eta - y_\eta z_\xi), & \zeta_y &= J(x_\eta z_\xi - x_\xi z_\eta), & \zeta_z &= J(x_\xi y_\eta - x_\eta y_\xi) \end{aligned}$$

- ژاکوبین تبدیل

$$\begin{aligned} J &= \frac{\text{area of computational space}}{\text{area of physical space}} = \frac{\partial(\xi, \eta, \zeta)}{\partial(x, y, z)} = \begin{vmatrix} \xi_x & \xi_y & \xi_z \\ \eta_x & \eta_y & \eta_z \\ \zeta_x & \zeta_y & \zeta_z \end{vmatrix} \\ &= \frac{1}{J^{-1}} = \frac{1}{\frac{\partial(x, y, z)}{\partial(\xi, \eta, \zeta)}} = \frac{1}{\begin{vmatrix} x_\xi & x_\eta & x_\zeta \\ y_\xi & y_\eta & y_\zeta \\ z_\xi & z_\eta & z_\zeta \end{vmatrix}} = \frac{1}{x_\xi(y_\eta z_\zeta - y_\zeta z_\eta) + x_\eta(y_\zeta z_\xi - y_\xi z_\zeta) + x_\zeta(y_\xi z_\eta - y_\eta z_\xi)} \end{aligned}$$

دسته‌بندی معادلات حاکم و انتقال

• انتقال مختصات (شبکه دوبعدی و متحرک)

- مشتق زنجیره‌ای

$$\begin{cases} \tau = t \\ \xi = \xi(x, y, t) \\ \eta = \eta(x, y, t) \end{cases} \rightarrow \begin{cases} \frac{\partial}{\partial t} = \frac{\partial}{\partial \tau} + \xi_t \frac{\partial}{\partial \xi} + \eta_t \frac{\partial}{\partial \eta} \\ \frac{\partial}{\partial x} = 0 + \xi_x \frac{\partial}{\partial \xi} + \eta_x \frac{\partial}{\partial \eta} \\ \frac{\partial}{\partial y} = 0 + \xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta} \end{cases}$$

$$\begin{cases} t = \tau \\ x = x(\xi, \eta, \tau) \\ y = y(\xi, \eta, \tau) \end{cases} \rightarrow \begin{cases} \frac{\partial}{\partial \tau} = \frac{\partial}{\partial t} + x_\tau \frac{\partial}{\partial x} + y_\tau \frac{\partial}{\partial y} \\ \frac{\partial}{\partial \xi} = 0 + x_\xi \frac{\partial}{\partial x} + y_\xi \frac{\partial}{\partial y} \\ \frac{\partial}{\partial \eta} = 0 + x_\eta \frac{\partial}{\partial x} + y_\eta \frac{\partial}{\partial y} \end{cases}$$

- محاسبه متریک‌های تبدیل

$$\begin{pmatrix} \partial_t \\ \partial_x \\ \partial_y \end{pmatrix} = \begin{pmatrix} 1 & \xi_t & \eta_t \\ 0 & \xi_x & \eta_x \\ 0 & \xi_y & \eta_y \end{pmatrix} \begin{pmatrix} \partial_\tau \\ \partial_\xi \\ \partial_\eta \end{pmatrix} \quad \& \quad \begin{pmatrix} \partial_\tau \\ \partial_\xi \\ \partial_\eta \end{pmatrix} = \begin{pmatrix} 1 & x_\tau & y_\tau \\ 0 & x_\xi & y_\xi \\ 0 & x_\eta & y_\eta \end{pmatrix} \begin{pmatrix} \partial_t \\ \partial_x \\ \partial_y \end{pmatrix}$$

دسته‌بندی معادلات حاکم و انتقال

- انتقال مختصات (شبکه دوبعدی و متحرک)

$$\begin{pmatrix} 1 & \xi_t & \eta_t \\ 0 & \xi_x & \eta_x \\ 0 & \xi_y & \eta_y \end{pmatrix} = \begin{pmatrix} 1 & x_\tau & y_\tau \\ 0 & x_\xi & y_\xi \\ 0 & x_\eta & y_\eta \end{pmatrix}^{-1} = J \begin{pmatrix} x_\xi y_\eta - y_\xi x_\eta & -x_\tau y_\eta + y_\tau x_\eta & x_\tau y_\xi - y_\tau x_\xi \\ 0 & y_\eta & -y_\xi \\ 0 & -x_\eta & x_\xi \end{pmatrix}$$

- متریک‌ها و ژاکوبین تبدیل

$$\xi_x = J y_\eta \quad \& \quad \eta_x = -J y_\xi \quad \& \quad \xi_y = -J x_\eta \quad \& \quad \eta_y = J x_\xi \quad \& \quad J^{-1} = x_\xi y_\eta - y_\xi x_\eta$$

$$\xi_t = (-x_\tau y_\eta + y_\tau x_\eta) J = -x_\tau \xi_x - y_\tau \xi_y \quad \& \quad \eta_t = (x_\tau y_\xi - y_\tau x_\xi) J = -x_\tau \eta_x - y_\tau \eta_y$$

- سرعت نقاط شبکه

$$x_\tau = \frac{\partial x}{\partial \tau} \approx \frac{x^{n+1} - x^n}{\Delta \tau} = \text{grid velocity}$$

دسته‌بندی معادلات حاکم و انتقال

- انتقال مختصات (شبکه دوبعدی و متحرک)

$$Q_t + F_x + G_y = 0 \rightarrow Q_\tau + \xi_t Q_\xi + \eta_t Q_\eta + \xi_x F_\xi + \eta_x F_\eta + \xi_y G_\xi + \eta_y G_\eta = 0$$

$$\frac{\xi_x F_\xi}{J} = \left(\frac{\xi_x F}{J} \right)_\xi - F \left(\frac{\xi_x}{J} \right)_\xi \quad \& \quad \frac{\xi_y G_\xi}{J} = \left(\frac{\xi_y G}{J} \right)_\xi - G \left(\frac{\xi_y}{J} \right)_\xi$$

$$\frac{\eta_x F_\eta}{J} = \left(\frac{\eta_x F}{J} \right)_\eta - F \left(\frac{\eta_x}{J} \right)_\eta \quad \& \quad \frac{\eta_y G_\eta}{J} = \left(\frac{\eta_y G}{J} \right)_\eta - G \left(\frac{\eta_y}{J} \right)_\eta$$

$$\frac{Q_\tau}{J} = \left(\frac{Q}{J} \right)_\tau - Q \left(\frac{1}{J} \right)_\tau \quad \& \quad \frac{\xi_t Q_\xi}{J} = \left(\frac{\xi_t Q}{J} \right)_\xi - Q \left(\frac{\xi_t}{J} \right)_\xi \quad \& \quad \frac{\eta_t Q_\eta}{J} = \left(\frac{\eta_t Q}{J} \right)_\eta - Q \left(\frac{\eta_t}{J} \right)_\eta$$

$$\begin{aligned} -Q \left\{ \left(\frac{1}{J} \right)_t + \left(\frac{\xi_t}{J} \right)_\xi + \left(\frac{\eta_t}{J} \right)_\eta \right\} + \left(\frac{Q}{J} \right)_\tau + \left(\frac{\xi_t Q}{J} \right)_\xi + \left(\frac{\xi_x F}{J} \right)_\xi + \left(\frac{\eta_x F}{J} \right)_\eta - F \left\{ \left(\frac{\xi_x}{J} \right)_\xi + \left(\frac{\eta_x}{J} \right)_\eta \right\} \\ + \left(\frac{\eta_t Q}{J} \right)_\eta + \left(\frac{\xi_y G}{J} \right)_\xi + \left(\frac{\eta_y G}{J} \right)_\eta - G \left\{ \left(\frac{\xi_y}{J} \right)_\xi + \left(\frac{\eta_y}{J} \right)_\eta \right\} = 0 \end{aligned}$$

دسته‌بندی معادلات حاکم و انتقال

- انتقال مختصات (شبکه دوبعدی و متحرک)

$$T_1 = \left(\frac{1}{J} \right)_t + \left(\frac{\xi_t}{J} \right)_\xi + \left(\frac{\eta_t}{J} \right)_\eta = 0, \quad T_2 = \left(\frac{\xi_x}{J} \right)_\xi + \left(\frac{\eta_x}{J} \right)_\eta = 0, \quad T_3 = \left(\frac{\xi_y}{J} \right)_\xi + \left(\frac{\eta_y}{J} \right)_\eta = 0$$

$$\left(\frac{Q}{J} \right)_\tau + \left(\frac{\xi_t Q}{J} \right)_\xi + \left(\frac{\xi_x F}{J} \right)_\xi + \left(\frac{\eta_x F}{J} \right)_\eta + \left(\frac{\eta_t Q}{J} \right)_\eta + \left(\frac{\xi_y G}{J} \right)_\xi + \left(\frac{\eta_y G}{J} \right)_\eta = 0$$

$$\left(\frac{Q}{J} \right)_\tau + \left(\frac{\xi_t Q}{J} + \frac{\xi_x F}{J} + \frac{\xi_y G}{J} \right)_\xi + \left(\frac{\eta_t Q}{J} + \frac{\eta_x F}{J} + \frac{\eta_y G}{J} \right)_\eta = 0$$

$$\tilde{Q}_\tau + \tilde{F}_\xi + \tilde{G}_\eta = 0 \quad \& \quad \tilde{Q} = \frac{1}{J} \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ E_t \end{pmatrix}, \quad \tilde{F} = \frac{1}{J} \begin{pmatrix} \rho U \\ \rho u U + \xi_x p \\ \rho v U + \xi_y p \\ (E_t + p)U - \xi_t p \end{pmatrix}, \quad \tilde{G} = \frac{1}{J} \begin{pmatrix} \rho V \\ \rho u V + \eta_x p \\ \rho v V + \eta_y p \\ (E_t + p)V - \eta_t p \end{pmatrix}$$

$$\begin{cases} U = \xi_t + \xi_x u + \xi_y v \\ V = \eta_t + \eta_x u + \eta_y v \end{cases} \quad \text{Contravariant velocity components}$$

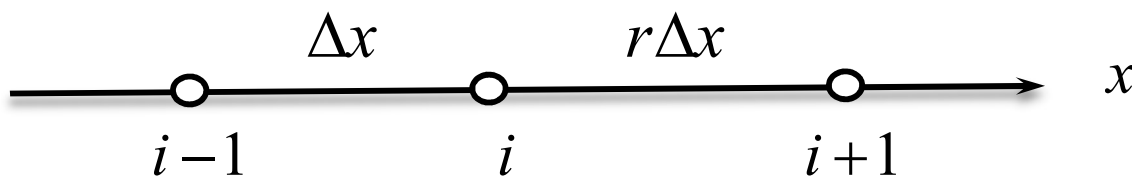
دسته‌بندی معادلات حاکم و انتقال

- انتقال مختصات

- نحوه محاسبه متریک‌های تبدیل

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \xi} \frac{\partial \xi}{\partial x} + \frac{\partial u}{\partial \eta} \frac{\partial \eta}{\partial x} \quad \text{or} \quad u_x = u_\xi \xi_x + u_\eta \eta_x$$

$$\eta = y \rightarrow \eta_x = 0 \rightarrow u_x = u_\xi \xi_x = u_\xi \frac{1}{x_\xi}$$



Physical domain



Computational domain

دسته‌بندی معادلات حاکم و انتقال

- انتقال مختصات

- نحوه محاسبه متریک‌های تبدیل

$$u_x|_{\text{num}} = u_\xi \xi_x|_{\text{num}} = u_\xi \frac{1}{x_\xi}|_{\text{num}} = \frac{u_{i+1} - u_{i-1}}{2\Delta\xi} \frac{1}{(x_{i+1} - x_{i-1}) / (2\Delta\xi)} = \frac{u_{i+1} - u_{i-1}}{x_{i+1} - x_{i-1}}$$

$$\begin{cases} u_{i+1} = u_i + \Delta\xi u_\xi + \frac{(\Delta\xi)^2}{2} u_{\xi\xi} + \frac{(\Delta\xi)^3}{6} u_{\xi\xi\xi} + \dots \\ u_{i-1} = u_i - \Delta\xi u_\xi + \frac{(\Delta\xi)^2}{2} u_{\xi\xi} - \frac{(\Delta\xi)^3}{6} u_{\xi\xi\xi} + \dots \end{cases}$$

$$u_{i+1} - u_{i-1} = 2\Delta\xi u_\xi + \frac{(\Delta\xi)^3}{3} u_{\xi\xi\xi} + \dots \quad \& \quad x_{i+1} - x_{i-1} = 2\Delta\xi x_\xi + \frac{(\Delta\xi)^3}{3} x_{\xi\xi\xi} + \dots$$

دسته‌بندی معادلات حاکم و انتقال

• انتقال مختصات

- نحوه محاسبه متریک‌های تبدیل

$$\begin{aligned}\left. \frac{\partial u}{\partial x} \right|_{\text{num}} &= \frac{u_{i+1} - u_{i-1}}{x_{i+1} - x_{i-1}} = \frac{u_{\xi} \left(1 + \frac{(\Delta \xi)^2}{6} \frac{u_{\xi\xi\xi}}{u_{\xi}} + \dots \right)}{x_{\xi} \left(1 + \frac{(\Delta \xi)^2}{6} \frac{x_{\xi\xi\xi}}{x_{\xi}} + \dots \right)} \\ &= \frac{u_{\xi}}{x_{\xi}} \left(1 + \frac{(\Delta \xi)^2}{6} \frac{u_{\xi\xi\xi}}{u_{\xi}} + \dots \right) \left(1 - \frac{(\Delta \xi)^2}{6} \frac{x_{\xi\xi\xi}}{x_{\xi}} + \dots \right) \\ &= \frac{u_{\xi}}{x_{\xi}} \left\{ 1 + \frac{(\Delta \xi)^2}{6} \left(\frac{u_{\xi\xi\xi}}{u_{\xi}} - \frac{x_{\xi\xi\xi}}{x_{\xi}} \right) + \dots \right\}\end{aligned}$$

$$u_{\xi} = u_x x_{\xi} \quad \& \quad u_{\xi\xi} = u_{xx} x_{\xi}^2 + u_x x_{\xi\xi} \quad \& \quad u_{\xi\xi\xi} = x_{\xi}^3 u_{xxx} + 3x_{\xi} x_{\xi\xi} u_{xx} + x_{\xi\xi\xi} u_x$$

دسته‌بندی معادلات حاکم و انتقال

- انتقال مختصات

- نحوه محاسبه متریک‌های تبدیل

$$\begin{aligned} u_x|_{\text{num}} &= u_x \left\{ 1 + \frac{(\Delta\xi)^2}{6} \left(\frac{x_\xi^3 u_{xxx} + 3x_\xi x_{\xi\xi} u_{xx} + x_{\xi\xi\xi} u_x}{x_\xi u_x} - \frac{x_{\xi\xi\xi}}{x_\xi} \right) + \dots \right\} \\ &= u_x + \frac{(\Delta\xi)^2}{6} \left\{ x_\xi^2 u_{xxx} + 3x_{\xi\xi} u_{xx} + \frac{x_{\xi\xi\xi}}{x_\xi} u_x - \frac{x_{\xi\xi\xi}}{x_\xi} u_x \right\} + \dots \\ &= u_x + \frac{(\Delta\xi)^2}{6} \left\{ x_\xi^2 u_{xxx} + 3x_{\xi\xi} u_{xx} \right\} + \dots \end{aligned}$$

$$x_\xi = \frac{x_{i+1} - x_{i-1}}{2\Delta\xi} + o(\Delta\xi)^2 = \frac{1}{2}(1+r) \frac{\Delta x}{\Delta\xi} + o(\Delta\xi)^2$$

$$x_{\xi\xi} = \frac{x_{i+1} - 2x_i + x_{i-1}}{(\Delta\xi)^2} + o(\Delta\xi)^2 = (r-1) \frac{\Delta x}{(\Delta\xi)^2} + o(\Delta\xi)^2$$

دسته‌بندی معادلات حاکم و انتقال

- انتقال مختصات

- نحوه محاسبه متریک‌های تبدیل

$$u_x|_{\text{num}} = u_x + \left\{ \frac{1}{6} \left(\frac{1+r}{2} \right)^2 (\Delta x)^2 u_{xxx} + \frac{1}{2} (r-1) \Delta x u_{xx} \right\} + \dots$$

uniform grid :

$$r = 1 \rightarrow u_x|_{\text{num}} \sim o(\Delta x)^2$$

nonuniform grid :

$$r \gg 1 \rightarrow x_{\xi\xi} \sim o(\Delta x) \quad \& \quad u_x|_{\text{num}} \sim o(\Delta x)$$

$$r-1 \sim o(\Delta x) \quad \text{or} \quad r \sim 1+o(\Delta x) \rightarrow \frac{r-1}{2} \Delta x \sim o(\Delta x)^2 \quad \& \quad u_x|_{\text{num}} \sim o(\Delta x)^2$$

دسته‌بندی معادلات حاکم و انتقال

- انتقال مختصات

- نحوه محاسبه متریک‌های تبدیل

$$u_x|_{\text{num}} = u_\xi \xi_x|_{\text{num}} = u_\xi \frac{1}{x_\xi}|_{\text{num}} = \frac{u_{i+1} - u_{i-1}}{2\Delta\xi} \frac{1}{x_\xi} = \frac{u_\xi + \frac{(\Delta\xi)^2}{6} u_{\xi\xi\xi} + \dots}{x_\xi}$$

$$u_\xi = u_x x_\xi \quad \& \quad u_{\xi\xi} = u_{xx} x_\xi^2 + u_x x_{\xi\xi} \quad \& \quad u_{\xi\xi\xi} = x_\xi^3 u_{xxx} + 3x_\xi x_{\xi\xi} u_{xx} + x_{\xi\xi\xi} u_x$$

$$u_x|_{\text{num}} = u_x + \frac{(\Delta\xi)^2}{6} \left(x_\xi^2 u_{xxx} + 3x_{\xi\xi} u_{xx} + \frac{x_{\xi\xi\xi}}{x_\xi} u_x \right) + \dots$$

- کاهش دقت حل عددی ناشی از عبارات اضافی هنگام تعیین متریک‌های تبدیل به صورت تحلیلی

- حذف عبارات اضافی در خطای برشی و افزایش دقت حل هنگام تعیین متریک‌های تبدیل به صورت عددی

دسته‌بندی معادلات حاکم و انتقال

- انتقال مختصات

- نحوه محاسبه متریک‌های تبدیل

$$J^{-1}u_x = J^{-1}(u_\xi \xi_x + u_\eta \eta_x) = u_\xi y_\eta - u_\eta y_\xi \quad \text{nonconservative form}$$

$$u = \text{const.} \quad \rightarrow \quad J^{-1}u_x = 0$$

$$J^{-1}u_x = u_\xi y_\eta - u_\eta y_\xi = (u y_\eta)_\xi - (u y_\xi)_\eta \quad \text{conservative form}$$

$$J^{-1}u_x \Big|_{\text{num}} = \frac{1}{2} \left\{ (u y_\eta)_{i+1,j} - (u y_\eta)_{i-1,j} - (u y_\xi)_{i,j+1} + (u y_\xi)_{i,j-1} \right\} + 0 \left[(\Delta \xi)^2, (\Delta \eta)^2 \right] \quad \& \quad \Delta \xi = \Delta \eta = 1$$

$$u = \text{const.} \quad \rightarrow \quad J^{-1}u_x \neq 0 \quad \text{when calculating the metrics analytically}$$

$$J^{-1}u_x \Big|_{\text{num}} = \frac{1}{2} \left\{ u_{i+1,j} (y_{i+1,j+1} - y_{i+1,j-1}) - u_{i-1,j} (y_{i-1,j+1} - y_{i-1,j-1}) - u_{i,j+1} (y_{i+1,j+1} - y_{i-1,j+1}) + u_{i,j-1} (y_{i+1,j-1} - y_{i-1,j-1}) \right\}$$

$$u = \text{const.} \quad \rightarrow \quad J^{-1}u_x = 0 \quad \rightarrow \quad \text{the metrics should be calculated numerically}$$

دسته‌بندی معادلات حاکم و انتقال

- دسته‌بندی معادلات حاکم

The general quasi-linear 1st-order system:

$$A_{ij} \frac{\partial u_j}{\partial x} + B_{ij} \frac{\partial u_j}{\partial y} = C_i, \quad i, j = 1, 2, 3, \dots, n$$

$$u_j(x, y) = \text{dependent variables} \quad \& \quad A = A(x, y, \vec{u}) \quad \& \quad B = B(x, y, \vec{u}) \quad \& \quad C = C(x, y, \vec{u})$$

$$\begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}_x + \begin{pmatrix} B_{11} & B_{12} & B_{13} \\ B_{21} & B_{22} & B_{23} \\ B_{31} & B_{32} & B_{33} \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}_y = \begin{pmatrix} C_1 \\ C_2 \\ C_3 \end{pmatrix}$$

- عدم امکان تعیین اطلاعات مربوط به تغییرات هر u_j در یک جهت مشخص

- استفاده از ترکیب معادلات برای این منظور

دسته‌بندی معادلات حاکم و انتقال

- دسته‌بندی معادلات حاکم

$$L_i \left(A_{ij} \frac{\partial u_j}{\partial x} + B_{ij} \frac{\partial u_j}{\partial y} \right) = L_i C_i \quad \& \quad \vec{L} = \vec{L}(x, y, \vec{u})$$

- بیان رابطه‌ای مابین مشتقات تمامی u_j ها در یک جهت مشخص (α, β)

(ترکیب سه معادله به یک معادله)

$$\begin{cases} L_i A_{ij} = m_j \alpha \\ L_i B_{ij} = m_j \beta \end{cases} \rightarrow m_j \left(\alpha \frac{\partial u_j}{\partial x} + \beta \frac{\partial u_j}{\partial y} \right) = L_j C_j$$

- معرفی منحنی مشخصه (x, y) توسط میدان برداری (α, β)

$$\begin{cases} x = x(s) \\ y = y(s) \end{cases} \text{ characteristic curve} \rightarrow u_j = u_j(x, y) = u_j(x(s), y(s))$$

$$\frac{du_j}{ds} = \frac{\partial u_j}{\partial x} \frac{dx}{ds} + \frac{\partial u_j}{\partial y} \frac{dy}{ds} = \alpha \frac{\partial u_j}{\partial x} + \beta \frac{\partial u_j}{\partial y} \quad \& \quad \begin{cases} \alpha = \frac{dx}{ds} \\ \beta = \frac{dy}{ds} \end{cases} \quad \text{مشتق کلی } u_j \text{ در جهت } (\alpha, \beta)$$

$$m_j \left(\alpha \frac{\partial u_j}{\partial x} + \beta \frac{\partial u_j}{\partial y} \right) = L_j C_j \rightarrow m_j \frac{du_j}{ds} = L_j C_j \quad \text{ODE} \quad \text{فرم مشخصه‌ای معادلات}$$

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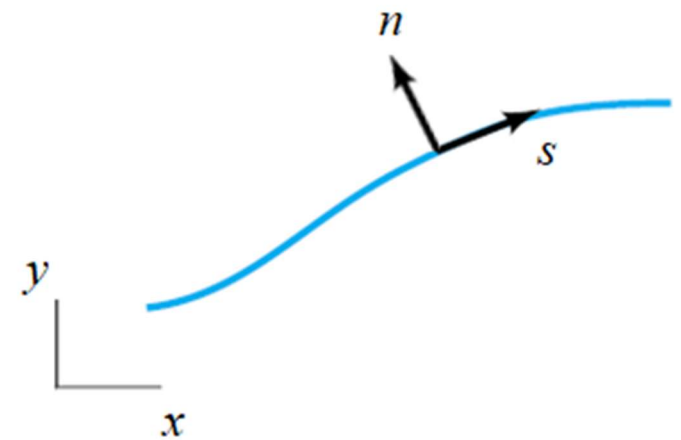
- شرایط تبدیل PDE به ODE

$$\begin{cases} L_i A_{ij} = m_j \alpha \\ L_i B_{ij} = m_j \beta \end{cases} \rightarrow \begin{cases} \beta L_i A_{ij} = m_j \alpha \beta \\ \alpha L_i B_{ij} = m_j \alpha \beta \end{cases} \rightarrow \beta L_i A_{ij} = \alpha L_i B_{ij}$$

$$L_i (\beta A_{ij} - \alpha B_{ij}) = 0 \rightarrow L_i \left(\frac{dy}{ds} A_{ij} - \frac{dx}{ds} B_{ij} \right) = 0$$

$$\hat{n} = n_x \hat{i} + n_y \hat{j} \rightarrow \begin{cases} \frac{dx}{dn} = -\frac{dy}{ds} = -\sin \theta \\ \frac{dy}{dn} = \frac{dx}{ds} = \cos \theta \end{cases} \rightarrow -\frac{n_x}{n_y} = \frac{dy}{dx} = \frac{\beta}{\alpha} = \lambda$$

$$L_i \left(\frac{dy}{ds} A_{ij} - \frac{dx}{ds} B_{ij} \right) = 0 \rightarrow L_i \left(\frac{dx}{dn} A_{ij} + \frac{dy}{dn} B_{ij} \right) = L_i (n_x A_{ij} + n_y B_{ij}) = 0$$



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- شرط وجود جواب برای منحنی‌های مشخصه

$$|T| = |n_x A_{ij} + n_y B_{ij}| = |n_x [A] + n_y [B]| = 0$$

- حالت خاص

$$A_{ij} = \delta_{ij} \rightarrow [A] = I \quad \text{unit matrix}$$

$$A_{ij} \frac{\partial u_j}{\partial x} + B_{ij} \frac{\partial u_j}{\partial y} = C_i \rightarrow \delta_{ij} \frac{\partial u_j}{\partial x} + B_{ij} \frac{\partial u_j}{\partial y} = C_i \rightarrow \frac{\partial u_i}{\partial x} + B_{ij} \frac{\partial u_j}{\partial y} = C_i$$

$$|T| = |n_x A_{ij} + n_y B_{ij}| = n_y \left| \frac{n_x}{n_y} \delta_{ij} + B_{ij} \right| = n_y |-\lambda \delta_{ij} + B_{ij}| = n_y |B - \lambda I| = 0$$

$$\begin{cases} L_i A_{ij} = m_j \alpha \\ L_i B_{ij} = m_j \beta \end{cases} \rightarrow \begin{cases} L_i \delta_{ij} = m_j \alpha \\ L_i B_{ij} = m_j \beta \end{cases} \rightarrow \begin{cases} L_j = m_j \alpha \\ L_i B_{ij} = m_j \beta \end{cases} \rightarrow L_i B_{ij} = \frac{\beta}{\alpha} L_j = \lambda L_j$$

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• دسته‌بندی معادلات حاکم

- مثال ۱

$$\nabla^2 \phi = \phi_{xx} + \phi_{yy} = 0 \quad \text{Laplace equation}$$

$$\begin{cases} \phi_x = u \\ \phi_y = v \end{cases} \rightarrow u_x + v_y = 0$$

$$\begin{cases} (\phi_x)_y = u_y \\ (\phi_y)_x = v_x \end{cases} \rightarrow v_x - u_y = 0$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}_x + \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}_y = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$T = n_x [A] + n_y [B] = n_x \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + n_y \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} n_x & n_y \\ -n_y & n_x \end{pmatrix}$$

$$\det T = \begin{vmatrix} n_x & n_y \\ -n_y & n_x \end{vmatrix} = n_x^2 + n_y^2 = 0 \rightarrow \left(\frac{n_x}{n_y} \right)^2 + 1 = 0 \rightarrow \left(\frac{n_y}{n_x} \right) = \pm i \quad \text{elliptic system}$$

دسته بندی معادلات حاکم و انتقال

2D steady inviscid incompressible flow :

$$\begin{cases} u_x + v_y = 0 \\ uu_x + vu_y + p_x = 0 \\ uv_x + vv_y + p_y = 0 \end{cases} \rightarrow A Q_x + B Q_y = 0$$

• دسته بندی معادلات حاکم

- مثال ۲

$$Q = \begin{pmatrix} u \\ v \\ p \end{pmatrix} \quad \& \quad A = \begin{pmatrix} 1 & 0 & 0 \\ u & 0 & 1 \\ 0 & u & 0 \end{pmatrix} \quad \& \quad B = \begin{pmatrix} 0 & 1 & 0 \\ v & 0 & 0 \\ 0 & v & 1 \end{pmatrix}$$

$$T = n_x [A] + n_y [B] = n_x \begin{pmatrix} 1 & 0 & 0 \\ u & 0 & 1 \\ 0 & u & 0 \end{pmatrix} + n_y \begin{pmatrix} 0 & 1 & 0 \\ v & 0 & 0 \\ 0 & v & 1 \end{pmatrix} = \begin{pmatrix} n_x & n_y & 0 \\ un_x + vn_y & 0 & n_x \\ 0 & un_x + vn_y & n_y \end{pmatrix}$$

$$\det T = n_x [-n_x (un_x + vn_y)] - n_y [n_y (un_x + vn_y)] = -(un_x + vn_y)(n_x^2 + n_y^2) = 0$$

$$-\frac{1}{n_x^3} \left(u + v \frac{n_y}{n_x} \right) \left(\left(\frac{n_y}{n_x} \right)^2 + 1 \right) = 0 \rightarrow \begin{cases} \frac{n_y}{n_x} = \pm i & \text{elliptic} \\ \frac{n_y}{n_x} = -\frac{u}{v} & \text{hyperbolic} \end{cases}$$

mixed hyperbolic/elliptic system

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1D unsteady inviscid compressible flow :

$$\frac{\partial Q}{\partial t} + \frac{\partial F_i}{\partial x} = 0, \quad Q = \begin{pmatrix} \rho \\ \rho u \\ E_t \end{pmatrix}, \quad F_i = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ (E_t + p)u \end{pmatrix}, \quad E_t = \rho \left(e + \frac{1}{2} u^2 \right) = \frac{p}{\gamma - 1} + \frac{1}{2} \rho u^2$$

• دسته‌بندی معادلات حاکم

- مثال ۳

$$\begin{cases} \rho_t + u \rho_x + \rho u_x = 0 \\ u_t + u u_x + \frac{1}{\rho} p_x = 0 \\ p_t + u p_x + \rho a^2 u_x = 0 \end{cases} \quad \text{primitive form} \quad \rightarrow \quad \frac{\partial \tilde{Q}}{\partial t} + \tilde{A} \frac{\partial \tilde{Q}}{\partial x} = 0, \quad \tilde{Q} = \begin{pmatrix} \rho \\ u \\ p \end{pmatrix} \quad \& \quad \tilde{A} = \begin{pmatrix} u & \rho & 0 \\ 0 & u & \frac{1}{\rho} \\ 0 & \rho a^2 & u \end{pmatrix}$$

$$\det T = |\tilde{A} - \lambda I| = \begin{vmatrix} u - \lambda & \rho & 0 \\ 0 & u - \lambda & \frac{1}{\rho} \\ 0 & \rho a^2 & u - \lambda \end{vmatrix} = (u - \lambda)(u - \lambda + a)(u - \lambda - a) = 0 \quad \rightarrow \quad \begin{cases} \lambda_1 = \frac{dx}{dt} \Big|_1 = u \\ \lambda_1 = \frac{dx}{dt} \Big|_2 = u - a \\ \lambda_1 = \frac{dx}{dt} \Big|_3 = u + a \end{cases} \quad \text{hyperbolic system}$$

موفق باشید.